

# LIFE OF A STAR

Lightweight stars live the longest – up to **10 trillion** years.

For most of its life, a star is powered by the nuclear fusion of hydrogen into helium. When all the hydrogen is used up, a lightweight star puffs off its outer layers, as a glowing planetary nebula. However, a heavier star can fuse a whole sequence of elements, turning into a red supergiant before exploding as a supernova.

## DATA FILE



**LIGHTEST STAR:** GJ 1245C, 0.07 of the Sun's weight



**HEAVIEST STAR:** R136a1, 315 times heavier than the Sun



**COOLEST STAR:** 2MASS J0523-1403, 1,800°C (3,270°F) at the surface



**HOTTEST STAR:** WR 102, 210,000°C (378,030°F) at the surface

What is space?

At an age of less than 10 million years, the star Betelgeuse is nearing the end of its life.